

# ANAND VYVASWATH KALVA

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## EDUCATION

### Carnegie Mellon University

May 2027

Master of Science, Mechanical Engineering - Research | Current GPA: 4.0/4.0

Pittsburgh, PA

Coursework: Modern Control Theory, Robot Dynamics and Analysis, ML and AI, Optimal Control and RL (In progress)

### Indian Institute of Technology

July 2024

Bachelor of Technology, Mechanical Engineering | GPA: 9.0/10

Indore, India

Relevant Coursework: Spacecraft Attitude Dynamics, Instrumentation & Control

## EXPERIENCE

### Carnegie Mellon University

Oct 2025 – Present

Graduate Researcher (Advisor: Prof. Hartmut Geyer)

Pittsburgh, PA

- Investigating transfer of locomotion control from high-level brain policies to low-level spinal controllers.
- Building and simulating musculoskeletal locomotion models in Python to study control and movement behavior.

### Carnegie Mellon University

Sept 2025 – Dec 2025

Course Assistant - Mechanical Systems Experimentation (24-452)

Pittsburgh, PA

- Conducted laboratory sessions and office hours for dynamic systems and controls experiments, including system identification, frequency response and PID control.
- Graded lab reports and provided feedback on experimental analysis.

### Indian Institute of Science

Jul 2024 – Mar 2025

Research Intern

Bengaluru, India

- Redesigned and optimized a modular articulated robot, enabling reliable locomotion in inclined pipes.
- Modeled and simulated robot dynamics, producing simulations that guided development of control strategies.

### Zebu Intelligent Systems

May 2023 – Jul 2023

Mechanical Intern

Hyderabad, India

- Designed and analyzed structural components for a surveillance helicopter, contributing to a working prototype.
- Designed the swashplate mechanism, analyzed transmission issues, and integrated Pixhawk Cube with ArduPilot for flight readiness.

## PROJECTS

### Gesture Controlled SMA Actuated Robotic Hand | Under Prof. I. A. Palani, IIT Indore

Aug 2023 - Dec 2023

- Developed a robotic hand actuated by Shape Memory Alloy (SMA) springs, achieving lightweight, silent and compact actuation.
- Developed a machine-learning-based gesture recognition interface achieving > 95 % real-time classification accuracy.
- Innovation led to a granted patent: "A Gesture-controlled SMA-actuated Robotic Gripper System and Method Thereof" (Patent No. 568766, July 2025).

### Synthetic Jet Impingement Cooling | Fluid Mechanics Lab, IIT Indore

Mar 2022 - Sep 2022

- Investigated heat transfer characteristics of high-aspect-ratio elliptical synthetic jets through thermal imaging, providing insights applicable to electronics cooling advancements.
- Processed and analyzed data on MATLAB, which culminated in a paper presented at the FMFP 2022 Conference.

## TECHNICAL SKILLS

**Programming:** Python, MATLAB

**Frameworks:** Gymnasium, PyTorch, ROS 2

**Technical:** Mujoco, SolidWorks, Fusion 360, Ansys Mechanical, ADAMS, Onshape

**Other:** Linux, Git, Arduino IDE,  $\LaTeX$

## PATENTS

### A Gesture-controlled SMA-actuated Robotic Gripper System and Method Thereof

Patent No. 568766, Grant Date: 17th July 2025

### Robotic Joint Apparatus with Multiple Degrees of Freedom and Method Thereof

Application No. 202541061762, Filed Date: 11th July 2025